

Sebastian Salinas

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EDUCATION

Texas Tech University, Lubbock, Texas

Bachelor of Science in Biochemistry

Summa Cum Laude | Minor: Biology

Coursework: Genetics, Biochemistry I-III, Biophysical Chemistry, Physical Chemistry I-II, Immunology, Organic Chemistry I-II, Microbiology, English 1 & 2

GPA: 4.0

May 2025

CURRENT RESEARCH

J20 Alzheimer's Mouse Model Lab | PI: Dr. Josh Lawrence, TTUHSC *May 2025 – Present*
Research Technician

Investigate the effect of epigenetic and metabolic interventions on neurodegeneration in the J20 transgenic Alzheimer's mouse model. All-trans retinoic acid (ATRA), a vitamin A derivative, and the HDAC inhibitor vorinostat have demonstrated neuroprotective effects. These compounds halt and/or reverse neurological deterioration in mammalian neuronal tissue through mechanisms that remain under active investigation. Research projects focus on dietary vitamin A supplementation in the form of ATRA and pharmacological HDAC inhibition with vorinostat and determining their effects on neurophysiology, excitability, electrophysiology, behavior, morphology, and biochemistry. Current work integrates qRT-PCR, Western blot, histology, microscopy, behavior, patch clamp, and dietary and pharmacological interventions to investigate how ATRA and vorinostat modulate Alzheimer's-related phenotypes. Supported by a Culture of Care Award from the Garrison Institute on Aging. Presented findings at SFN 2025.

RESEARCH TRAINING

Keyel Lab, Texas Tech University, Lubbock, Texas

November 2022 – May 2025

Undergraduate Researcher | Advisor: Dr. Peter Keyel

- Examined how cholesterol and sphingolipid-rich plasma membrane microdomains in *Leishmania major* influence susceptibility to pore-forming toxins.
- Achieved LC50 values for streptolysin O (SLO), a cholesterol-dependent cytolysin (CDC), as well as preformed binding assays on non-CDC variants of the toxins.
- Assessed surface coat proteins as modulators of cytolytic susceptibility, with lipophosphoglycan and 14-methylated sterols protecting against SLO-mediated pore formation.

- These findings advance understanding of membrane repair and have implications for host-pathogen interactions, wound repair, and membrane biophysics. Presented findings at 2 ASCB meetings.

Cardiovascular Mechanics Lab, Lubbock, Texas

December 2021 – May 2022

Volunteer Researcher

- Assisted in designing cardiovascular heart valves under Dr. Zhaoming He.
- Conducted advanced literary analysis and meta-analyses using online research databases and Blood flow analysis via MATLAB.

NASA Space and Life Sciences Institute, Houston, Texas

May 2022 – August 2022

Research Intern

- Observed studies on the effects of lunar dust and simulants on lung epithelial cells in Dr. Jeeverajan's lab, employing fluorescence microscopy and chemical modeling software.
- Developed strategies to mitigate contact with lunar materials.

Johnson Space Center, NASA, Houston, Texas *April 2018 – May 2020, May 2023 – August 2023*

Technical Intern

- 18th Annual Space Life Sciences Summer Institute intern. Technical internship and project design in Energy Systems design.
- Use of computer design software and CAD programs (such as Inventor).

Tutor | Dept. of Biological Sciences, Texas Tech University, Lubbock, Texas *January 2022 – May 2022*

- Tutor for introductory biology lab, covering molecular biology, genetics, classical and molecular evolutionary biology, and ecology and environmental biology.

TEACHING

Teaching Assistant, Confocal Microscopy – GSBS Masters Program, Dept. of Pharmacology and Neuroscience, TTUHSC

Sept 2026 – Present

Graduate Teaching Assistant to Dr. Josh Lawrence

- Assist Dr. Josh Lawrence with masters-level confocal microscopy instruction for GSBS students.
- Guide a small cohort (4 in first session, 2 additional expected) through hands-on scope time: multichannel imaging, acquisition, and ZEN software.
- Support sample preparation and quantitative image analysis; deliver introductory lecture on confocal principles.

PUBLICATIONS

1. CS Haram, **SJ Salinas**, C Wilson, SW Sheikh, K Zhang, & PA Keyel. (2026) (Publication), *Functional Dissection of Leishmania major Membrane Components in Resistance to Cholesterol-Dependent Cytolysins*. Toxins. 2026; 18(1):46.
<https://doi.org/10.3390/toxins18010046>

IN PREPARATION

1. Chaitanya S. Haram, **Sebastian Salinas**, Salma Waheed Sheikh, Kai Zhang, & Peter A. Keyel. *Leishmanial GP63 acts as a protease for the small pore forming toxin aerolysin*. doi: <https://doi.org/10.64898/2026.07.29.741567>
1. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Retinoic acid and HDAC Inhibition reprogram oxidative stress responses via context-dependent antioxidant and transcriptional mechanisms in cultured neurons*. Working draft.
2. Baker, A., **Salinas, S.**, Lie, P., Smith, S.C., Skawratananond, S., Bose, C., & Lawrence, J.J. *KCNIP1 at the Convergence of Nutritional Deficiency, Ion Channel Dysfunction, and Alzheimer's Disease*. Working draft.

ACCEPTED ABSTRACTS

1. Lie, P.E., Baker, A., Self, D., Khanna, N., Zhang, C., **Salinas, S.**, Smith, S.C., Skawratananond, S., Kisby, B.R., Shanmugam, S., Ponomarev, I., Telias, M., & Lawrence, J.J. *A Dopaminergic Hippocampal Interneuron Associated with Cognitive Severity Links Retinoic Acid Signaling, and Synaptic Scaffolding Failure in the J20 Alzheimer's Mouse Model*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.
2. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of Dietary Vitamin A in Dentate Gyrus Granule Cell Excitability in a Mouse Model of Alzheimer's Disease*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.
3. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A-Dependent Modulation of Behavioral and Biochemical Profiles in an Alzheimer's Disease Mouse Model*. Alzheimer's Association International Conference, London, United Kingdom, July 12th-15th, 2026.
4. **Salinas, S.**, Baker, A., Smith, S.C., Strickland, J., Bose, C., & Lawrence, J.J. *Long-Term Vitamin A Supplementation Decelerates Cognitive Deficits and Anxiety-Like Phenotypes in the J20 Alzheimer's Mouse Model*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.
5. Baker, A., Smith, S.C., **Salinas, S.**, Bose, C., Skawratananond, S., Tanner, N., & Lawrence, J.J. *Behavioral and Cellular Effects of Vitamin A Deficiency in a Mouse*

Model of Alzheimer's Disease. Alzheimer's Disease International Conference, London, United Kingdom, July 12th-15th, 2026.

POSTER PRESENTATIONS

1. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A supplementation modulates behavioral and biochemical profiles in a mouse model of Alzheimer's disease*. CTNT/GIA, TTUHSC, April 2026
2. Lie, P.E., Baker, A., Khanna, N., Self, D., Chakroborty, A., Smith, S.C., **Salinas, S.**, Skawratananond, S., Kisby, B.R., Shanmugan, M., Telias, M., Ponomarev, I., & Lawrence, J.J. *Computational ethology characterizes behavioral inflexibility and excitatory/inhibitory transcriptional dysregulation in vitamin A deficient Alzheimer's disease model mice*. CTNT/GIA, TTUHSC, April 2026
3. Luong, K., Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A Supplementation Reduces Anxiety and Improves Spatial Cognition in the J20 Mouse Model of Alzheimer's Disease*. CTNT/GIA, TTUHSC, April 2026
4. Kim, A., Zhang, C., **Salinas, S.**, Hindle, A., Baker, A., Smith, S.C., Singh, S.P., Lawrence, J.J., & Bose, C. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in hippocampal neurons*. Undergraduate Research Conference, Texas Tech University, April 2026
5. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A on dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. 9th Annual Symposium on Brain Health, GIA, TTUHSC, April 2026
6. Baker, A., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A Supplementation Reduces Anxiety and Improves Spatial Cognition in the J20 Mouse Model of Alzheimer's Disease*. Texas Alzheimer's Research and Care Consortium, Austin, TX, January 2026
7. Baker, A., Smith, S.C., **Salinas, S.**, Bose, C., Skawratananond, S., Tanner, N., & Lawrence, J.J. *Behavioral and cellular effects of vitamin A deficiency in a mouse model of Alzheimer's disease*. Alzheimer's Disease International Conference, London, United Kingdom, July 12th-15th, 2026
8. Lie, P.E., Baker, A., Khanna, N., Self, D., Chakroborty, A., Smith, S.C., **Salinas, S.**, Skawratananond, S., Kisby, B.R., Shanmugan, A., Hindle, M., Telias, M., Ponomarev, I., & Lawrence, J.J. *Machine learning demonstrates vitamin A-mediated partial rescue of behavioral phenotype in an Alzheimer's disease model mouse*. Student Research Week, TTUHSC, February 2026
9. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Sex differences in dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. Student Research Week, TTUHSC, February 2026

10. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A on dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. First Annual Meeting on Human Molecular Aging, Innovation Hub at Research Park, April 2026
11. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A in dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. 38th Annual Student Research Week, TTUHSC, February 2026
12. Kim, A., Zhang, C., **Salinas, S.**, Hindle, A., Baker, A., Smith, S.C., Singh, S.P., Lawrence, J.J., & Bose, C. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in hippocampal neurons*. 38th Annual Student Research Week, TTUHSC, February 2026
13. Kim, A., Zhang, C., **Salinas, S.**, Baker, A., Smith, S.C., Singh, S.P., Bose, C., & Lawrence, J.J. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in the hippocampus of a mouse model of Alzheimer's disease*. Student Research Week, Texas Tech University, April 2026
14. **S. Salinas**, C. Haram, K. Zhang, & P. Keyel (2024) (Poster). *14-methylated sterols protect the human pathogen Leishmania major against membrane damage by cholesterol-dependent cytolysins*. Mol. Biol Cell 36. Abstract #2057. American Society of Cell Biologist (ASCB) Cell Bio 2024
15. C. Haram, R. Keane, **S. Salinas**, S. Waheed, K. Zhang, P. Keyel (2024). *Leishmania major is an unconventional system to study membrane damage by bacterial toxins*. Mol Biol Cell 36. Abstract #2944
16. **S. Salinas**, C. Haram, K. Zhang, & P. Keyel (2023) (Poster). *Non calcium ions regulate membrane repair in human pathogen Leishmania major*. Mol. Biol Cell 35. Abstract #2900. ASCB Cell Bio 2023
17. C. Haram, R. Keane, **S. Salinas**, K. Zhang, P. Keyel (2023). *Lipophosphoglycan shields Leishmanial GPI anchored proteins from the pore forming toxin aerolysin*. Mol Biol Cell 35. Abstract #1561
18. **S. Salinas**, C. Haram, S. Waheed, K. Zhang, P. Keyel (2024) (Talk). *Leishmania is a system to study cell survival*. TrUE Symposium, Abstract #4d1. Texas Tech Center for Transformative Undergraduate Experiences.
19. **S. Salinas**, C. Haram, K. Zhang, P. Keyel (2023) (Poster). *Detailing mechanism behind Membrane Repair in Leishmaniasis*. TTU Undergraduate Research Conference, Abstract #A174. Discoveries to Impact 2023.

SKILLS

Laboratory Techniques:

- 4 years of experience: PCR, Immunohistochemistry, Confocal Microscopy, Widefield fluorescence microscopy.
- 3 years of experience: Flow Cytometry, Western Blotting, Leishmania Promastigote Cell Culture, SDS-PAGE.

- 2 years of experience: High Performance Liquid Chromatography, Column chromatography, Mass spectroscopy.
- 1 year of experience: Mammalian Cell culture, Transformation, Plasmid Isolation, Protein Purification, Mouse colony maintenance, Rodent handling, Rodent Tissue collection.

Imaging & Analysis:

- Volume rendering, segmentation, tracing, digital reconstruction, 3D visualization, quantitative analysis of neuronal and subcellular structures.

Software & Tools:

- ImageJ, Imaris, fluorescence quantification pipelines, MATLAB, FlowJo, Benchling, GraphPad Prism, Adobe Photoshop, Biorender. Blood flow analysis via MATLAB.

Techniques:

- DNA/RNA extraction, qRT-PCR, spectrophotometry, gel electrophoresis, ELISA, protein biochemistry, sterile technique, tissue processing, cryosectioning, histology, immunofluorescence staining, cell culture, CRISPR-Cas9, Solution and buffer preparation, chemical handling, sample preparation, lab equipment operation and maintenance, mechanical and chemical tissue disaggregation.

Research:

- Experimental design, cohort selection, blinding, randomization, data analysis, data management, scientific writing, lab meetings, literature reviews, hypothesis formulation, record-keeping, manuscript writing.

Certifications:

- CITI human subjects, CITI animal use, OSHA bloodborne pathogens.

HONORS AND LEADERSHIP

Tech10 recipient, Texas Tech University

May 2025

President, Raider Recycling

Aug 2023 - May 2025

- Reinitiated recycling collection in residence halls across the Texas Tech Campus.
- Lead a team of 8 officers to enhance sustainability projects and organize recycling initiatives at Texas Tech.

Outstanding Student Leadership Award, Center for Integrated STEM and Education Research

April 2025

Mentor, Honors College Mentor Program, Texas Tech University

March 2022 - March 2025

Extra Kilometer Award, Center for Integrated STEM and Education Research

April 2024

Texas Tech Association of Biologists, Active Member

Aug 2024 - May 2025

- Participate in organizational volunteering activities as well as social activities.

CISER (Center for Integration of STEM Education & Research), Active Member

Mar 2021 - May 2025

- Conduct science outreach for middle and elementary school students.

REFERENCES

Dr. J. Josh Lawrence, Ph.D. Dept. of Pharmacology and Neuroscience, Texas Tech University Health Sciences Center, John.Lawrence@ttuhsc.edu, 806-743-2656

Dr. Chhanda Bose, Ph.D. Dept. of Pharmacology and Neuroscience, Texas Tech University Health Sciences Center, chhanda.bose@ttuhsc.edu, 806-743-2801

Dr. Peter Keyel, Ph.D. Dept. of Biological Sciences, Texas Tech University, peter.keyel@ttu.edu, 806-834-8241

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